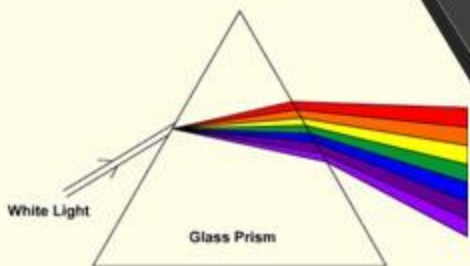


PHYSICS

MAGNETIC EFFECTS OF ELECTRIC CURRENT

MODULE - 16

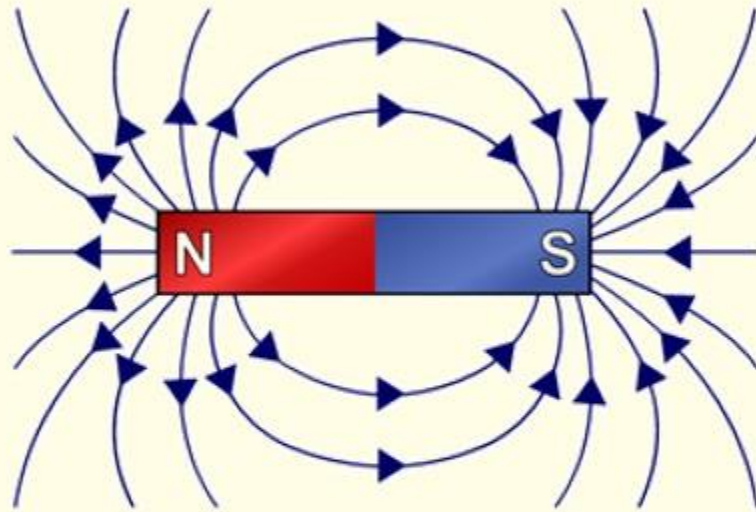


PROPERTIES OF MAGNET



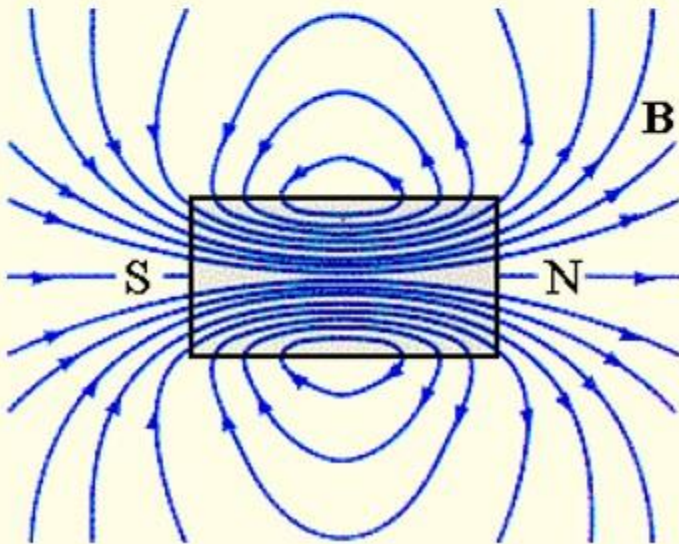
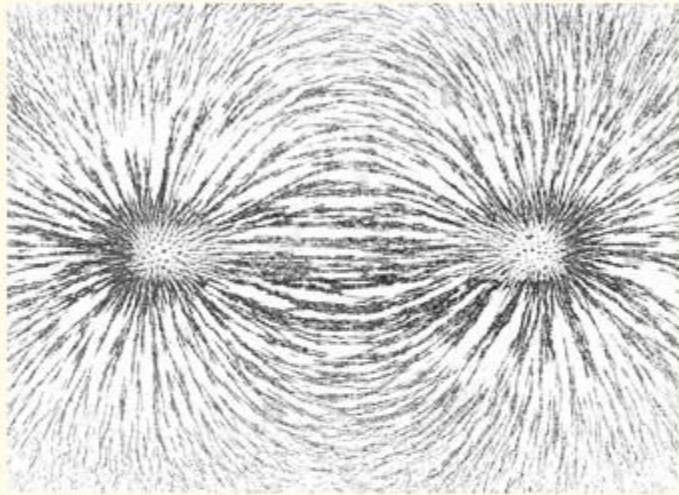
- A free suspended magnet always point towards north and south direction.
- The pole of a magnet which points toward north direction is called north pole or north seeking.
- The pole of a magnet which points toward south direction is called south pole or south seeking.
- Like poles of magnets repel each other while unlike poles of magnets attract each other.

MAGNETIC FIELD



- The influence of force surrounding a magnet is called magnetic field.
- In the magnetic field, the force exerted by a magnet can be detected using a compass or any other magnet.
- The imaginary lines of magnetic field around a magnet are called field line or field line of magnet.
- Magnetic field is a vector quantity, i.e. it has both direction and magnitude

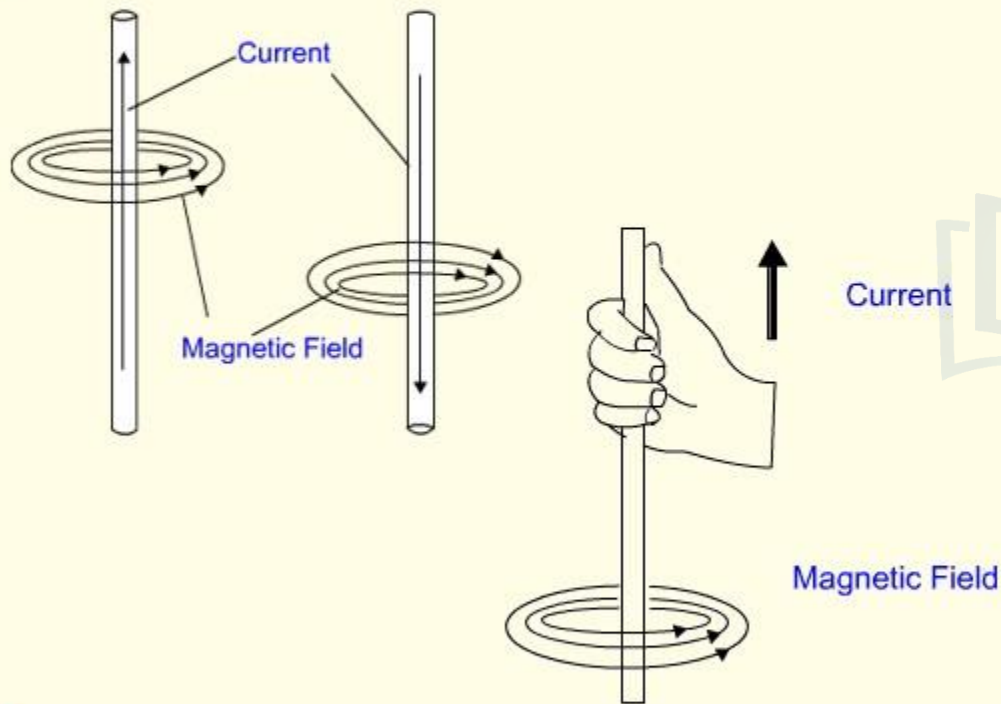
PROPERTIES OF MAGNETIC FIELD



- 1 Outside the magnet, the direction of magnetic field line is taken from north pole to South Pole. Inside the magnet, the direction of magnetic field line is taken from south pole to north pole.
- 2 The closeness of field lines shows the relative strength of magnetic field, i.e. closer lines show stronger magnetic field and vice-versa. Crowded field lines near the poles of magnet show more strength.
- 3 Magnetic field lines are always parallel to each other.
No two field lines cross each other.

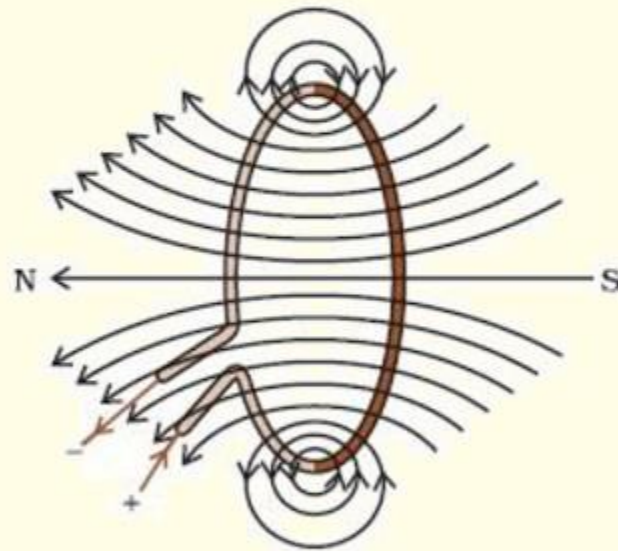


RIGHT HAND THUMB RULE OF A MAGNETIC FIELD DUE TO A CURRENT CARRYING CONDUCTOR



- A current carrying straight conductor has magnetic field in the form of concentric circles; around it.
- If a current carrying conductor is held by right hand; keeping the thumb straight and if the direction of electric current is in the direction of thumb, then the direction of wrapping of other fingers will show the direction of magnetic field.

MAGNETIC FIELD DUE TO CURRENT THROUGH A CIRCULAR LOOP



- A circular loop is made up of large number of very small straight wires.
- Each small section of current carrying wire contributes to magnetic field lines.
- At the centre of circular wire, field lines become straight and perpendicular to the plane of coil.
- Every section of the wire contributes to the magnetic field lines in the same direction within the loop.

Factors on which magnetic field produced at centre of a current carrying circular wire depends



The amount of current flowing through the wire

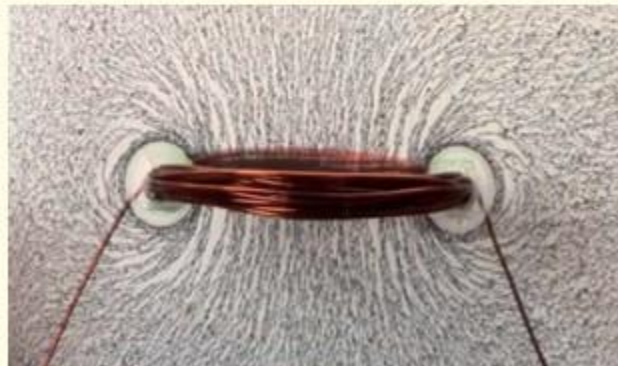
If current is increased then deflection of needle of magnetic compass placed inside wire also increases.

Number of turns in a circular wire

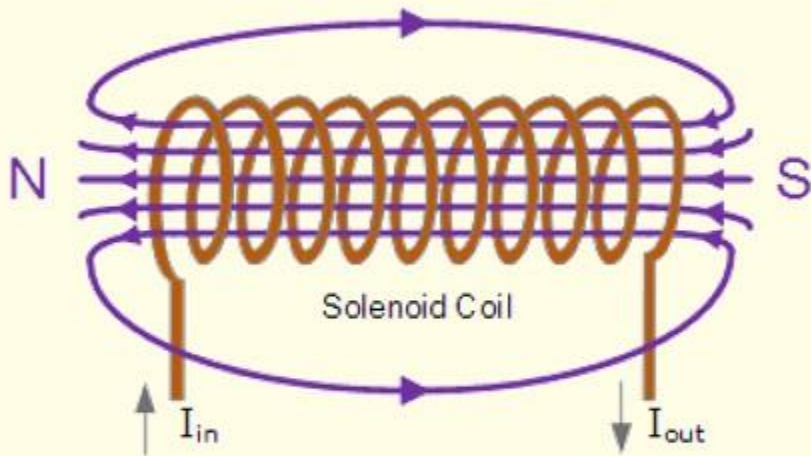
If number of turns in circular wire are increased, Magnetic field at the centre of loop increases.

Radius of a circular wire

Strength of magnetic field is inversely proportional to radius of circular wire.

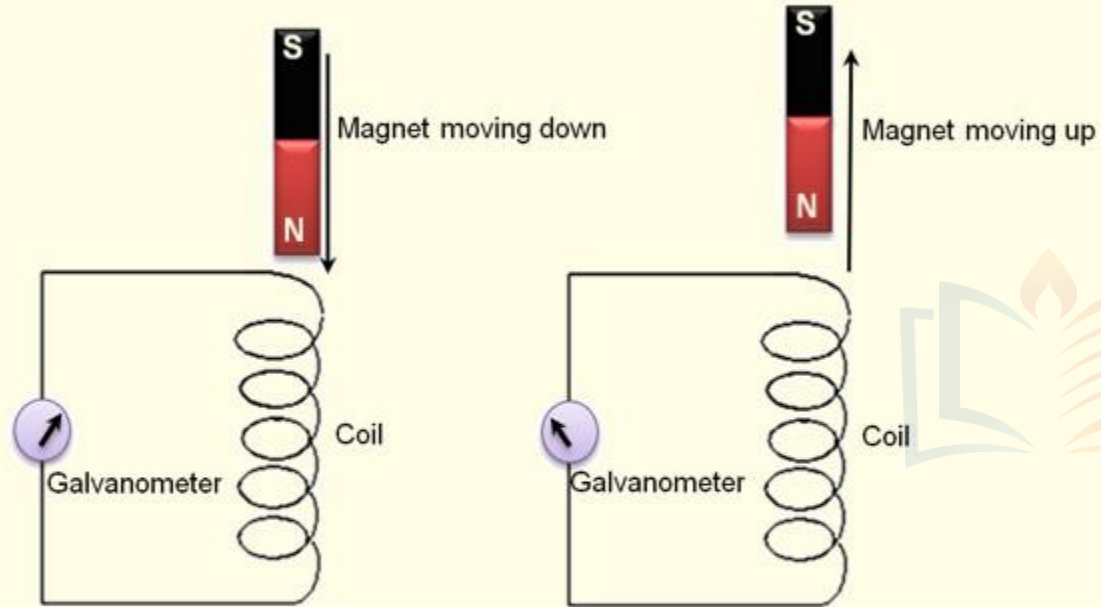


MAGNETIC FIELD DUE TO A CURRENT IN A SOLENOID



- A solenoid is a coil of many turns of an insulated copper wire closely wound in the shape of a tight spring.
- When an electric current flows through a solenoid, a magnetic field is set up around the solenoid similar to that of a bar magnet.
- One end of a solenoid acts as a north pole and the other as a south pole.
- Magnetic field inside a long solenoid decreases as we move towards the ends of the solenoid because magnetic field lines near the ends of the solenoid start spreading out.
- It can be used in a simple locking device, medical clamping equipment, an automotive gear box, and an air conditioning unit.

ELECTROMAGNETIC INDUCTION



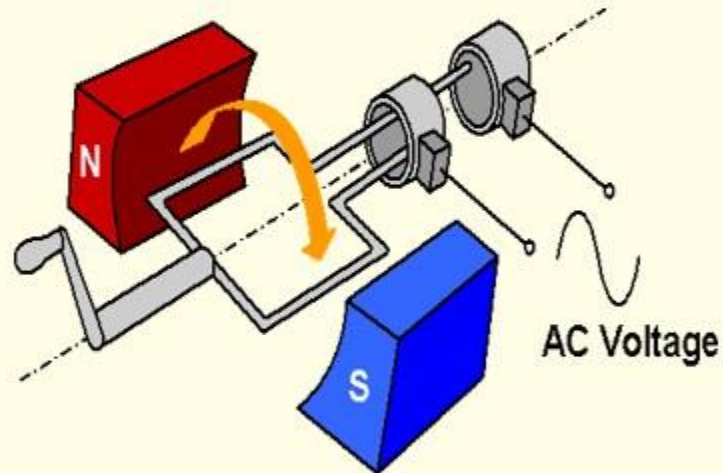
- Electromagnetic Induction was first discovered way back in the 1830's by Michael Faraday.
- A conductor placed in a changing magnetic field (or a conductor moving through a stationary magnetic field) causes the production of a voltage across the conductor.
- This process of electromagnetic induction, in turn, causes an electrical current - it is said to induce the current.

APPLICATIONS OF ELECTROMAGNETIC INDUCTION

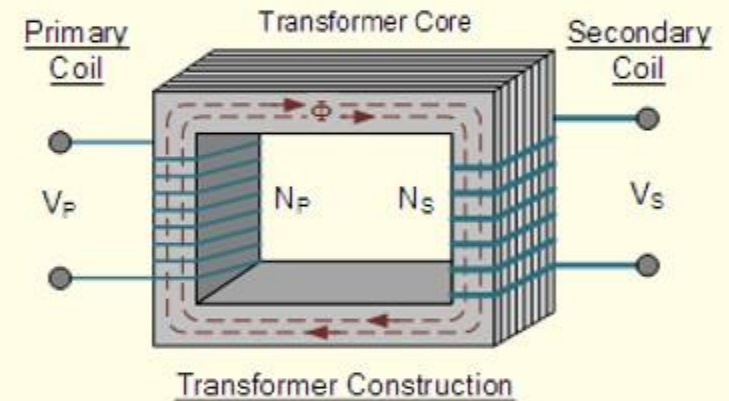
EDDY CURRENT



AN ELECTRIC GENERATOR



TRANSFORMERS



APPLICATIONS OF ELECTROMAGNETIC INDUCTION

EDDY CURRENT

An eddy current is a swirling current set up in a conductor in response to a changing magnetic field. The magnetic interaction between the applied field and the eddy currents acts to slow the wheels down.

AN ELECTRIC GENERATOR

When the device is used as a motor, a current is passed through the coil. The interaction of the magnetic field with the current causes the coil to spin. To use the device as a generator, the coil can be spun, inducing a current in the coil.

TRANSFORMERS

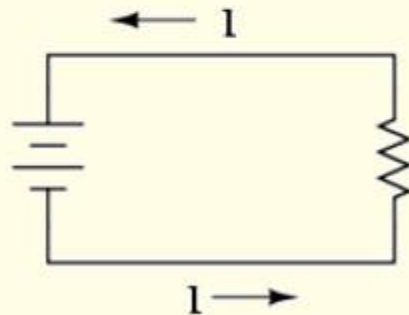
In a transformer, the two coils are usually wrapped around the same iron core. The coil that is connected to the AC source is known as the primary coil, while the coil in which voltage is induced is known as the secondary coil.



DIFFERENCE BETWEEN AC AND DC

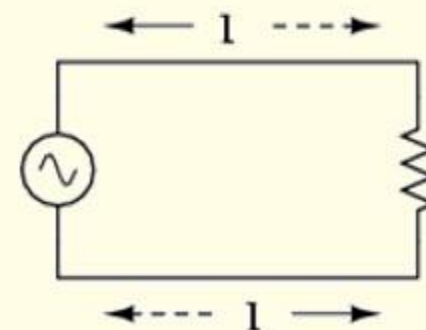
ALTERNATING CURRENT

- Current that switches direction of flow on regular time intervals
- Induced in a coil of wire turning in a magnetic field



DIRECT CURRENT

- Current that flows in only one direction through a circuit
- Created by a chemical reaction that produces a potential difference between the two terminals

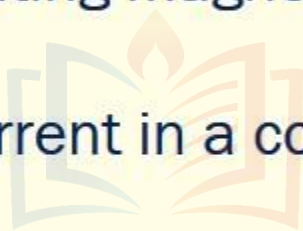




QUESTION 1

The phenomenon of electric induction is

- a) The process of charging a body
- b) The process of generating magnetic field lines due to current passing through a coil
- c) Producing induced current in a coil due to its relative motion between the magnet and the coil
- d) The process of rotating a coil of an electric motor



QUESTION 2

Whose magnetic field is like a bar magnet

- a) Current carrying wire
- b) Current carrying loop
- c) Current carrying solenoid
- d) Current carrying rectangular loop



QUESTION 3

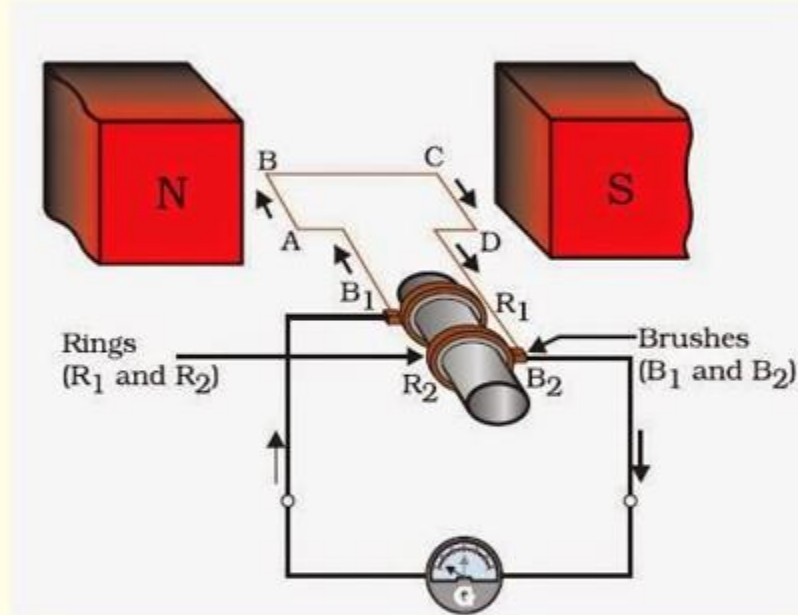
How many times an Alternating current with frequency of 50Hz change its direction?

- a) 50 times
- b) 100 times
- c) 200 times
- d) 25 times

Learning Space

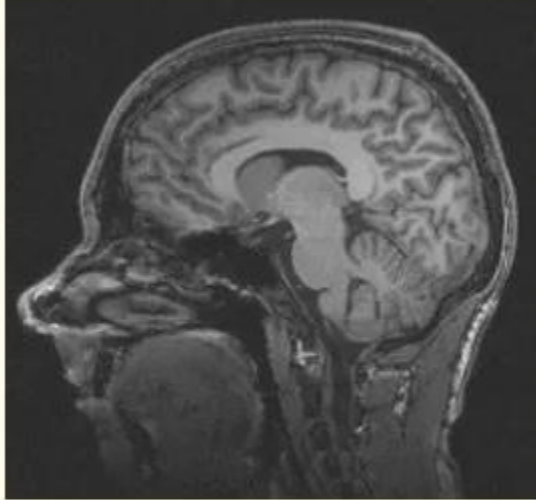


ELECTRIC GENERATOR



- An electric generator converts mechanical energy into electrical energy.
- when a coil of insulated copper wire is forced to rotate inside a magnetic field then electric current is induced.
- Every half rotation the polarity of the current changes and the device is called alternating current.
- The arrangement is also used to get a direct current.
- The direct current always flows in one direction, whereas the alternating current reverses its direction periodically.

MAGNETIC PROPERTIES OF QUANTUM PARTICLES : MRI



- Magnetic resonance imaging (MRI) makes use of the magnetic properties of the hydrogen nucleus (a single proton) present in water molecules, and therefore in all body tissues.
- The hydrogen nuclei behave like tiny compass needles aligned by a strong magnetic field of the scanner.
- The nuclei can be rotated using radio waves, and they subsequently oscillate in the magnetic field while returning to equilibrium.
- Simultaneously they emit a radio signal.
- MRI does not involve radioactivity or ionising radiation.
- Very detailed images can be made of soft tissues such as muscle and brain.
- MRI is consequently a very flexible technique that provides measures of both structure and function.

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SINCERE THANKS
FOR VIEWING THIS VIDEO

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